Offline
Calibration data
6 h
x(t)/x(6)
72 h
n units, m conditions
late label y ∈ {0, 1}
Posterior producer
p̂t ≈ Pr(y=1 | Ft)
p*
calibration at the stopping time: κ(τ)
Rule generation
DP for each μ (Thm 1)
family G = {πμ} of bands
fit-free: |p̂t − p*| ≥ m
Selection
fastest feasible rule π̂
α
π
risk + slack ≤ α;
else the endpoint assay
Certification
independent units
Thm 3: w.p. ≥ 1 − δ
new conditions
Thm 4: in expectation
output: band (at, bt)
Online
same producer
deploy the certified band
new unit
condition z
observe
x(tj)
posterior
p̂tj
p̂tj ∈ (atj, btj)?
no
stop at τ and decide
ŷ = 1{p̂τ ≥ p*}
yes: wait one step, pay λ(tj+1 − tj)
stop
bt
at
p̂t
τ
time
Thm 2 excess realized cost ≤ 2Lκ̄ + η, with κ̄ the calibration error of p̂ at the stopping time and η the discretization slack
Cor 1 |believed − realized cost| ≤ Lκ(τ) for every rule, so early-decision rules must be scored on realized cost